Newborn jaundice is a yellowing of a baby's skin and eyes. Newborn jaundice is very common and can occur when babies have a high level of bilirubin, a yellow pigment produced during normal breakdown of red blood cells. In older babies and adults, the liver processes bilirubin, which then passes it through the intestinal tract. However, a newborn's still-developing liver may not be mature enough to remove bilirubin. The good news is that in most cases, newborn jaundice goes away on its own as a baby's liver develops and as the baby begins to feed, which helps bilirubin pass through the body. In most cases, jaundice will disappear within 2 to 3 weeks. Jaundice that persists longer than 3 weeks may be a symptom of an underlying condition. Additionally, high levels of bilirubin can put a baby at risk for deafness, cerebral palsy, or other forms of brain damage. The American Academy of Pediatrics (AAP) recommends that all newborn babies be examined for jaundice before discharge from the hospital and again when the baby is between 3 and 5 days old.

## 1.1 Newborn jaundice:

Neonatal jaundice or neonatal hyperbilirubinemia results from elevated total serum bilirubin (TSB) and clinically manifests as yellowish discoloration of the skin, sclera, and mucous membrane. The term jaundice derives from the French word "jaune," which means yellow. It is the most commonly encountered medical problem in the first two weeks of life and a common cause of readmission to the hospital after birth.[1] Approximately 60% of term and 80% of preterm newborns develop clinical jaundice in the first week after birth.[2] In most cases, it is a mild, transient, and self-limiting condition and resolves without treatment referred to as "physiological jaundice." However, it is imperative to distinguish this from a more severe form called "pathological

jaundice." Failure to identify and treat this entity may result in bilirubin encephalopathy and associated neurological sequelae.

Unconjugated hyperbilirubinemia (UHB) is the cause of clinical jaundice in most neonates, but some infants with jaundice have conjugated hyperbilirubinemia (CHB), which is always pathological and signifies an underlying medical or surgical cause. The etiology of pathological UHB and CHB is vast and varied. Preterm infants and those born with congenital enzyme deficiencies are particularly prone to the harmful effects of unconjugated bilirubin on the central nervous system.[3][4] Severe hyperbilirubinemia can cause bilirubin-induced neurological dysfunction (BIND) and, if not treated adequately, may lead to acute and chronic bilirubin encephalopathy.[5] Phototherapy and exchange transfusions are the mainstay of treatment of UHB, and a subset of patients also respond to intravenous immunoglobulin (IVIG). Treatment of CHB is more complex and depends mainly on the etiology. Despite advances in care and management of hyperbilirubinemia, it remains a significant cause of morbidity and mortality.[6]

## 1.2 Two distinct types of Neonatal hyperbilirubinemia:

### 1.2.1 Unconjugated Hyperbilirubinemia(UHB) or Indirect Hyperbilirubinemia

Unconjugated hyperbilirubinemia is the more common type and is either physiological or pathological. Physiological jaundice accounts for 75% of neonatal hyperbilirubinemia and results from a physiological alteration in neonatal bilirubin metabolism. Healthy adults have a normal TSB level of less than 1mg/dl in contrast to neonates, where TSB levels are physiologically higher. Even in healthy full-term newborns, there is an increased bilirubin load owing to increased red blood cells (RBC) mass and a decreased RBC lifespan. Clearance of bilirubin is also compromised due to impaired activity of uridine diphosphate glucuronosyltransferase (UGT), the enzyme needed for bilirubin conjugation. The UGT enzyme in a newborn has an activity of about 1% of the adult

level.[7] Moreover, these infants also have increased enterohepatic circulation, further contributing to elevated TSB levels. Physiological jaundice typically appears after 24 hours of age, peaks at around 48-96 hours, and resolves by two to three weeks in full-term infants.[2]

Jaundice is considered pathological if it presents on the first day of life, TSB is more than the 95th centile for age based on age-specific bilirubin nomograms, levels rise by more than 5 mg/dL/day or more than 0.2 mg/dL/hour, or jaundice persists beyond 2 to 3 weeks in full-term infants.[8]

Based on the mechanism of bilirubin elevation, the etiology of unconjugated hyperbilirubinemia can be subdivided into the following three categories:

#### 1.2.1.1 Increased Bilirubin Production

Immune-mediated hemolysis - Includes blood group incompatibilities such as ABO and Rhesus incompatibility.

Non-immune mediated hemolysis - includes RBC membrane defects like hereditary spherocytosis and elliptocytosis; RBC enzyme defects like glucose-6-phosphate dehydrogenase (G6PD) deficiency; pyruvate kinase deficiency; sequestration like cephalohematoma, subgaleal hemorrhage, Intracranial hemorrhage; polycythemia, and sepsis.

#### 1.2.1.2 Decreased Bilirubin Clearance

Crigler-Najjar type I & II, and Gilbert syndrome.

### 1.2.2 Conjugated Hyperbilirubinemia(CHB) or Direct Hyperbilirubinemia

Conjugated hyperbilirubinemia, also referred to as neonatal cholestasis, is characterized by elevation of serum conjugated/direct) bilirubin (> 1.0 mg/dL) and is due to impaired hepatobiliary function. Distinguishing CHB from UHB is critical because cholestatic

jaundice/CHB is almost always pathologic and warrants prompt evaluation and treatment.[33]

The causes of neonatal cholestasis/CHB are extensive and can be classified into the following categories:

**Obstruction of biliary flow:** Biliary atresia, choledochal cysts, neonatal sclerosing cholangitis, neonatal cholelithiasis

**Infections:** CMV, HIV, rubella, herpes virus, syphilis, toxoplasmosis, urinary tract infection (UTI), septicemia

**Genetic causes**: Alagille syndrome, alpha-1 anti-trypsin deficiency, galactosemia, fructosemia, Tyrosinemia type 1, cystic fibrosis, progressive familial intrahepatic cholestasis (PFIC), Aagenaes syndrome, Dubin-Johnson syndrome, Bile acid synthesis disorders(BSAD)

**Miscellaneous:** Idiopathic neonatal hepatitis, parenteral nutrition induced cholestasis, gestational alloimmune liver disease/neonatal hemochromatosis, hypotension

## 1.3 Types of newborn jaundice:

### 1.3.1 physiological jaundice

- Not harmful
- very common in newborn
- seen during first 3-5 days of life
- disappear as liver matures

**For example:** baby has jaundice rate 12 and his age is 3 days

This baby has physiological jaundice and need only to breastfeeding

Figure **[1.1]** physiological jaundice

### 1.3.2 pathological jaundice:

- Harmful (occurs when the bilirubin levels are so high)
- Can affect brain cells, cause baby to be less active
- Can result in seizures, deafness, cerebral palsy, mental retardation

**For example**: baby has jaundice rate 18, his age is 18 days and his weight is 3450 gram

This baby is in serious condition and need to be in nursery

Figure **[1.2]** Pathological jaundice

## 1.4 Causes of newborn jaundice:

Babies at highest risk for developing newborn jaundice are:

- premature babies (babies born before 37 weeks' gestation)
- babies who aren't getting enough breast milk or formula, either because they're having a hard time feeding or because their mother's milk isn't in yet
- babies whose blood type isn't compatible with the blood type of their mother

A baby whose blood type isn't compatible with that of their mother can develop a buildup of antibodies that can destroy their red blood cells and cause a sudden rise in bilirubin levels.

Other causes of newborn jaundice include:

- bruising at birth or other internal bleeding
- liver problems
- an infection
- an enzyme deficiency
- an abnormality in your baby's red blood cells

## 1.5 Symptoms of newborn jaundice

The first sign of jaundice is a yellowing of a baby's skin and eyes. The yellowing may begin within 2 to 4 days after birth and may start in the face before spreading down across the body. Bilirubin levels typically peak between 3 to 7 days after birth. If a finger lightly pressed on a baby's skin causes that area of skin to become yellow, it's likely a sign of jaundice.

## 1.6 When to call a doctor

Most cases of jaundice are normal, but sometimes jaundice can indicate an underlying medical condition. Severe jaundice also increases the risk of bilirubin passing into the brain, which can cause permanent brain damage.

### Contact your doctor if you notice the following symptoms:

- The jaundice spreads or becomes more intense.
- Your baby develops a fever over 100°F (38°C).
- Your baby's yellow coloring deepens.

- Your baby feeds poorly, appears listless or lethargic, and makes high-pitched cries.

## 1.7 How is newborn jaundice diagnosed?

The hospital discharges most mothers and newborns within 72 hours of delivery. It's very important for parents to bring their babies in for a checkup a few days after birth because bilirubin levels peak between 3 to 7 days after birth. A distinct yellow coloring confirms that a baby has jaundice, but additional tests may be needed to determine the severity of the jaundice. Babies who develop jaundice in the first 24 hours of life should have bilirubin levels measured immediately, either through a skin test using jaundice meter or blood test.

We determine jaundice based on three major factors:

1. Weight

2. Age

3. Jaundice rate

**jaundiceasZaxmeter :**

Figure **[1.3]** jaundiceas Zaxmeter :

It is non-invasive method for screening babies for jaundice. The product is convenient, effective and proven, reducing the need for blood sampling and thus contributing to developmental care practice

But it is expensive to be in every home

The price is between 19000:60000 according to the accuracy

## So the major goals of our application are:

1. Every mother can have the application without paying anything

2. Be able to follow up the child's health condition without having to do a blood test from your baby

3. Caring for the health of the newborn baby

Most of applications in app store and google play aren't determine jaundice , only add jaundice rate , weight and age, and determine level of danger in jaundice